

INFLUENCE OF SHORT VS. LONG REPETITION SPRINT TRAINING ON SELECTED FITNESS COMPONENTS IN YOUNG SOCCER PLAYERS

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ABSTRACT

Meckel, Y, Gefen, Y, Nemet, D, and Eliakim, A. Influence of short vs. long repetition sprint training on selected fitness components in young soccer players. *J Strength Cond Res* 26(7): 1845–1851, 2012—The aim of this study was to compare the effect of short-sprint repetition and long-sprint repetition training (SST, LST), matched for total distance, on selected fitness components in young soccer players. Thirty young (14–15 years) soccer players were randomly assigned to either the short-sprint training group or long-sprint training group and completed 2 similar sets of fitness tests before and after 7 weeks of training. The 2 training programs consisted of SST (4–6 sets of 4 × 50-m all-out sprint) and LST (4–6 sets of 200-m run at 85% of maximum speed), each performed 3 times a week. Before training, there were no baseline between-group differences in predicted $\dot{V}O_2\text{max}$, standing long jump, 30-m sprint time, 4 × 10-m shuttle running time, and 250-m running time. Both training programs led to a significant improvement in $\dot{V}O_2\text{max}$ (predicted from the 20-m shuttle run, $p < 0.01$), with no between-group difference ($p = 0.14$). Both training programs also led to a significant improvement in the anaerobic fitness variables of 30-m sprint time ($p < 0.01$), 4 × 10-m shuttle running time ($p < 0.01$), and 250-m running time ($p < 0.01$), with no between-group differences. Neither of the training programs had a significant effect on standing long jump ($p = 0.21$). The study showed that long, near-maximal sprints, and short, all-out sprint training, matched for total distance, are equally effective in enhancing both the aerobic and anaerobic fitness of young soccer players. Therefore, to maintain a player's training interest and enthusiasm, coaches may alternate between these methods during the busy soccer season.

KEY WORDS speed, repetitions, interval, aerobic, adolescence

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INTRODUCTION

During a soccer game, elite level players cover 8–13 km at an average intensity close to the anaerobic threshold (7). The majority of the activity in the game is completed as low-intensity activities, such as walking, jogging, or slow running (18). The submaximal nature of this activity predominantly uses aerobic energy sources (1). However, within this endurance context, numerous explosive bursts of activity are required, including sprinting, jumping, turning, and tackling. Sprint-type activities account for approximately 8–12% of the total distance covered in the game, including mainly short (~15-m) and brief (~3-second) efforts (17). The maximal, or near maximal, nature of these activities uses anaerobic energy sources (1). For these reasons, soccer training should enhance both aerobic and anaerobic abilities.

In modern soccer, the high frequency of matches during the competitive season limits the number of training sessions devoted to fitness development. Thus, it would be valuable to identify a single training method that could efficiently improve both the aerobic and anaerobic abilities of the players. Various studies have shown that sprint training consisting of high-intensity short-term efforts of 5–30 seconds can improve anaerobic capabilities (4,6,14,16). In addition, it was found that sprint training can be effective in enhancing $\dot{V}O_2\text{max}$ and aerobic enzyme activity (4,6,14,20,21). These studies support the effectiveness of sprint or high-intensity interval training for enhancing both aerobic and anaerobic capacities. Furthermore, the intense muscle activity during sprint training may potentially be beneficial in improving muscular power and the ability to change direction or to tackle, as often required in soccer. Therefore, high-intensity intermittent-type training appears to be an appropriate exercise for improving the overall fitness requirements of soccer players. However, the specific sprint training method for soccer that would be most effective is not known. Specifically, whether short all-out sprint training is more effective than high-intensity long-sprint training is still unknown. To the authors' knowledge, a comparison between the effects of these 2 anaerobic-type training methods, matched for total distance, in young soccer players, was not performed.

Therefore, the aim of this study was to determine the effect of all-out short-sprint repetition training (SST) compared with high-intensity long-sprint repetition training (LST), matched for total distance and resting intervals, on the aerobic and anaerobic fitness of young soccer players. We hypothesized that compared with LST, SST would induce greater improvement in anaerobic fitness characteristics, but would lead to similar positive changes in aerobic fitness as well.

METHODS

Experimental Approach to the Problem

The study was conducted during the preparation period for the upcoming soccer season, immediately after 6 weeks of summer vacation. At this point, the players are usually in a relatively low physical fitness level and a large training-related improvement in fitness is expected. The study lasted 9 weeks and consisted of 1 week of pretesting, 7 weeks of specific training, and 1 week of posttesting. Because aerobic capacity is the major fitness component in soccer, the participants were randomly assigned to 1 of the 2 training groups after matching for aerobic fitness (based on predicted VO_2max determined from pretraining 20-m shuttle run test).

The 2 training programs consisted of SST (4–6 sets of $4 \times 50\text{-m}$ all-out sprint) and LST (4–6 sets of 200-m run at 85% of maximum speed), matched for total distance and total resting time. Each of these training protocols was performed 3 times a week. Because the study was conducted at the beginning of the training season and the players' physical condition was relatively low, training load was increased gradually throughout the 7 weeks of the study. To isolate the effect of the 2 training protocols, and to avoid any effect of other variables, no additional fitness training sessions (e.g., strength, speed, or aerobic) were performed during the 7 training weeks of the study.

Pretraining and posttraining aerobic fitness assessments were based on the maximal 20-m shuttle run aerobic power test. Pretraining and posttraining anaerobic fitness assessments included standing long jump, 30-m sprint time, $4 \times 10\text{-m}$ shuttle running time, and 250-m running time, which are commonly used in soccer training.

Subjects

Thirty young male trained soccer players (age 14.5 ± 0.6 years, body mass 57.2 ± 7.2 kg, height 167.1 ± 6.1 cm, body fat $8.8 \pm 3.8\%$), members of 2 teams from the first division youth league, participated in the study. The players had a similar training and playing background starting in the club's youth teams (around the age of 10). These teams were consistently ranked among the first 5 teams in the league. The players were informed of the experimental procedures and requirements before signing an informed consent. The study procedures, benefits, and risks were also explained to the parents and guardians before giving their written informed consent. All the procedures were conducted in accordance with the standards of the Institutional Ethical Committee.

Standard, calibrated scales and stadiometers were used to determine body weight and height. Skinfold measurement at 4 sites (triceps, biceps, subscapular, and suprailiac) was taken by an experienced technician.

Training Intervention

During the 7-week preseason training period, the participants trained $4 \text{ d} \cdot \text{wk}^{-1}$ with no games during the weekends. Out of these, 3 days were devoted to specifically designed training programs, with additional minor tactical and technical drills. The additional drills were similar in both training groups. No additional fitness training sessions (e.g., strength, speed, or aerobic) were performed during the 7 weeks of training. On the additional weekly training day, coaches focused on technical drills that were matched between groups in terms of type, time, and intensity. For complete matching in training load between groups, the participants were asked to avoid any leisure physical activity other than the soccer training. Table 1 describes the progress of the 2 training programs over the 7 weeks. Both training groups gradually increased training volume over the weeks in an attempt to avoid overtraining or injuries among the participants. For the SST group, the short-sprint training consisted of $4 \times 50\text{-m}$ all-out sprint repetitions for each set, with a 2-minute rest between each repetition (50 m), and a 4-minute rest between each set ($4 \times 50 \text{ m}$). The number of sets was increased gradually from 4 in the first and the second weeks to 5 in the third to fifth weeks and 6 in the final sixth and seventh weeks. Accordingly, total training distance increased from 800 to 1,200 m over the 7 weeks. For the LST group, the training consisted of long (200-m) repetitions (performed at 85% maximal speed based on each participant's best 100-m running time). Running pace was familiar to the participants from previous training. Five minutes of rest was given between each 200-m repetition. The number of repetitions was increased gradually from 4 in the first and second weeks to 5 in the third through fifth weeks and 6 in the final sixth and seventh weeks. Accordingly, the total distance had increased from 800 to 1,200 m over the 7 weeks. Overall, the 2 training groups had performed the same volume of work and were given the same total recovery time (15–25 minutes) over the 7 weeks of training (Table 1). The participants performed an active recovery between runs (walking around at a comfortable pace). All the training sessions were performed on a grass playing field with players wearing their soccer shoes. Training was performed during the late afternoon, 3–4 hours after lunch, with an average air temperature of about $24\text{--}26^\circ\text{C}$. The participants were instructed to drink 500 ml of water 30 minutes before each training session. None of the participants were taking any food supplements.

Testing Procedures

The participants performed 2 similar sets of tests before and after the 7-week training period. The first set of tests was conducted on 2 nonconsecutive days during the week before the beginning of the training program. The second set of tests was conducted under the same conditions during the week

TABLE 1. The training variables and their progression in the 2 training groups during the 9 weeks of training.*

Training group	Week number	Number of reps.	Total distance per session (m)	Total rest time per session (min)	Percent speed (%)
Pretests	1				
SST 4 × 50 m	2, 3	16	800	15	100
	4, 5, 6	20	1,000	20	
	7, 8	24	1,200	25	
Posttests	9				
Pretests	1				
LST 200 m	2, 3	4	800	15	85
	4, 5, 6	5	1,000	20	
	7, 8	6	1,200	25	
Posttests	9				

*LST = long-sprint repetition training; SST = short-sprint repetition training.

after the completion of the training program. Both sets of tests were performed using the same procedures, at the same time of the day, under the same environmental conditions, and with the same technician, who was blinded to the training-group affiliation. The participants were familiar with the testing procedures because they had routinely performed them in previous years. Before each set of tests, the participants performed a standard warm-up that included 8 minutes of jogging, followed by a 10-minute stretching exercise and two 30-m submaximal runs (at ~90% of maximal sprint speed). A 30-minute period separated the different tests on each day. All the running tests were performed at the same location and under similar environmental conditions as in the training sessions. Running times were recorded using a photoelectric cell timing system (Alge-Timing Electronic, Vienna, Austria) with an accuracy of 0.001 second linked to a digital chronoscope. A standing start, with the front foot placed 30 cm behind the timing lights, was used for all the sprints. Timing was initiated when the subject broke the light

beam. The rationale for testing selection was the application of the fitness components and movement type to the game of soccer. This selection is also consistent with other reports studying fitness of soccer players (23). Moreover, short-sprinting and jumping ability was shown to be significantly different in elite compared with that in average-level young soccer players (19). A high test-retest reliability (0.80–0.95), of the tests used in this study, was previously described (8,13,22,23).

On the first day of each testing set, the participants performed the following 3 tests:

Standing Long Jump—Power Test. Horizontal-jump length was measured by a maximum jump using a countermovement jump technique. The participants began in an erect standing position and moved into a semisquat position, leaning forward, before jumping horizontally as far as possible. All the jumps were performed in a long jump pit filled with soft sand. The participant used a vigorous double-arm swing as he

TABLE 2. The effect of the different training programs on selected aerobic and anaerobic performance indices.*

	Short-sprint training (n = 11)		Long-sprint training (n = 13)	
	Pre	Post	Pre	Post
$\dot{V}_{O_2\max}$ (ml·kg ⁻¹ ·min ⁻¹)	44.6 ± 5.3	47.5 ± 3.7†	44.9 ± 5.2	49.5 ± 5.4†
250-m Run (s)	39.82 ± 2.17	38.40 ± 2.27†	38.58 ± 2.48	37.51 ± 2.52†
30-m Run (s)	4.86 ± 0.19	4.74 ± 0.20†	4.82 ± 0.23	4.74 ± 0.22†
4 × 10-m Run (s)	9.83 ± 0.23	9.55 ± 0.33†	9.69 ± 0.34	9.59 ± 0.36†
Standing long jump (m)	2.13 ± 0.13	2.16 ± 0.17	2.10 ± 0.18	2.15 ± 0.16

*Values are given as mean ± SD.

†p < 0.01 for within-group change. There were no significant between-group differences.

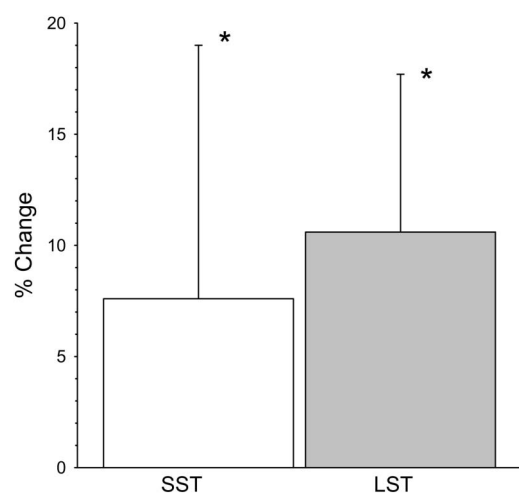


Figure 1. The effect of the different training programs on $\dot{V}O_{2\max}$ changes (predicted from the 20-m shuttle run test results). No significant between-group differences were found, $*p < 0.01$ for within-group changes.

jumped off the ground. Three trials were attempted, and the longest jump achieved was recorded. Jumping length was measured manually using a standard measuring tape.

The 4 × 10-m Run—Agility Test. The agility test used was the 4 × 10-m back and forth shuttle run. The participant began at the base line and ran at maximal speed to a marked line that was located 10 m ahead of him. He then turned and ran back to the baseline, turned again, and performed the same back and forth run. This procedure was performed twice, and the fastest time of the 2 trials was recorded. The participant rested for 5 minutes between the 2 runs. Each participant ran individually.

The 250-m Run—Anaerobic Endurance Test. The anaerobic endurance test consisted of a 250-m run. The run was performed on a 400-m lap track. The participants ran the 250 m in small homogenous groups of 3–4 players with similar capabilities (according to their personal records). Although each participant ran as fast as he could, each had the liberty to use his own tactics to produce the best possible result.

On the second day, the participants performed the following 2 tests:

The 30-m Run—Speed Test. The speed test consisted of a 30-m all-out sprint run. Two runs were completed, and the best time was recorded. The participants rested for 5 minutes between runs. The test was performed individually by each participant.

The 20-m Shuttle Run—Aerobic Power Test. The test consisted of shuttle runs at increasing speeds between 2 markers placed 20 m apart. A portable compact disc (Sony CFD-V7, Tokyo, Japan) dictated the test pace by emitting tones at appropriate intervals. The participant was required to be at one end of the 20-m course before the signal. A start speed of $8.5 \text{ km} \cdot \text{h}^{-1}$ was maintained for 1 minute, and thereafter, the speed was increased every minute by $0.5 \text{ km} \cdot \text{h}^{-1}$. The test score achieved was the number of 20-m laps completed before the subject either withdrew voluntarily from the test or failed to arrive within 3 m of

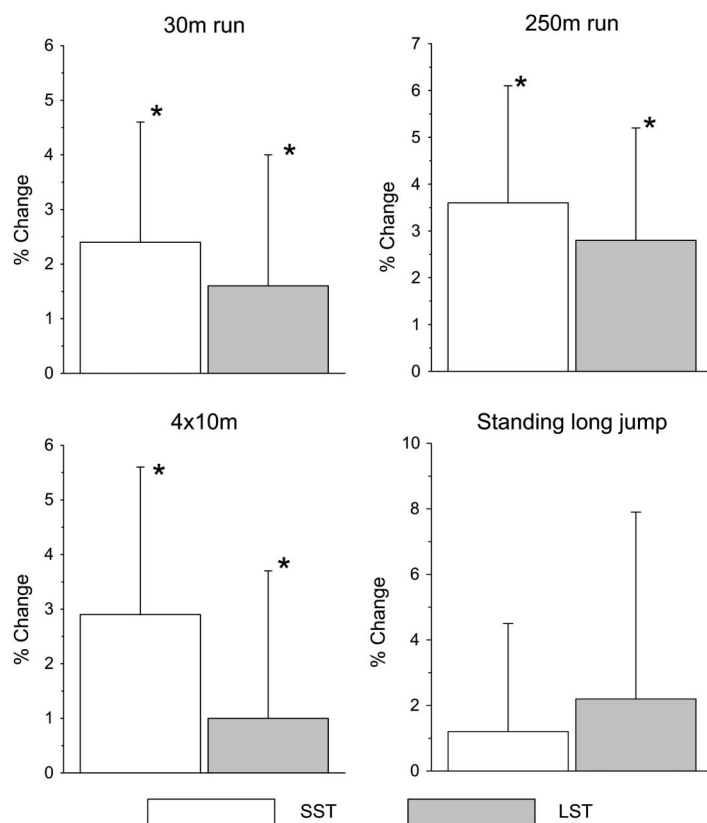


Figure 2. The effect of the different training programs on percent changes of selected anaerobic performance variables. There were no significant between-group differences, $*p < 0.01$ for within-group changes.

the end line on 2 consecutive tones. $\dot{V}O_2$ was derived by the formula: $Y = 6.0X - 24.4$, where Y equals the predicted $\dot{V}O_{2\max}$ and X equals the maximum speed achieved (22).

Statistical Analyses

A 2-way repeated-measure analysis of variance was used to compare fitness differences, with time serving as the within-group, and training regimen as the between-group factor. Data are presented as mean $\pm$ SD. Significance level was set at $p \leq 0.05$.

RESULTS

Six of the 30 players who started the study were excluded from the final analysis. The 20% drop-out rate occurred because of missed follow-up tests, injuries, or absence of $>10\%$ of the training sessions. Thus, only 24 participants (age 14.3 ± 0.5 years, body mass 56.5 ± 10.9 kg, height 166.1 ± 8.1 cm, body fat $9.3 \pm 4.8\%$) were included in the final analysis.

Changes in predicted $\dot{V}O_{2\max}$ values (based on the 20-m shuttle run test) are summarized in Table 2 and Figure 1. There were no baseline differences in the predicted $\dot{V}O_{2\max}$ before the training between the groups. Both training programs led to a significant improvement in the predicted $\dot{V}O_{2\max}$ ($p < 0.01$), with no between-group differences ($p = 0.14$).

Changes in the selected anaerobic performance variables are summarized in Table 2 and Figure 2. There were no baseline differences between the groups in the standing long jump, 30-m sprint run time, 4×10 -m shuttle run time, or 250-m run time before training. Both training programs led to a significant improvement in the 30-m sprint time ($p < 0.01$), 4×10 -m shuttle running time ($p < 0.01$), and 250-m running time ($p < 0.01$) but had no significant effect on standing long jump ($p = 0.21$). Changes in the 30-m sprint time ($p = 0.37$), 4×10 -m shuttle running time ($p = 0.11$), and the 250-m running time ($p = 0.39$) were slightly greater in the SST compared with that in the LST program, but these differences were not statistically significant.

DISCUSSION

The results of this study showed that both the SST and LST training programs led to a significant improvement in aerobic power, speed, agility, and anaerobic endurance, with no between-group differences. Neither training programs induced significant effects in jumping ability.

In agreement with our hypothesis, both LST and SST induced similar positive changes in aerobic power (10.7 and 7.5%, respectively). The changes in aerobic fitness after the LST are in line with those of previous studies that found a significant improvement in $\dot{V}O_{2\max}$ after 4–6 repeated 30-second efforts separated by a 4-minute recovery (2,4,9). Increases in oxidative enzyme activity have been implicated in mediating these responses. Moreover, it has been demonstrated that sprint interval training upregulates peroxisome proliferator-activated receptor-coactivator-1 α (PGC-1 α), a potent regulator of mitochondrial biogenesis

(3,10), which could be the underlying mechanism responsible for the aerobic adaptation. The significant improvement in aerobic fitness after the SST in this study is also in line with results of previous studies (6,11) reporting a significant increase in $\dot{V}O_{2\max}$ after short-sprint training. For example, Dawson et al. (6) found a significant improvement in $\dot{V}O_{2\max}$ of untrained young men after a training program that consisted of short (30–80 m) sprints performed 3 times per week. In another study, Ferrari Bravo et al. (8) found a significant improvement in $\dot{V}O_{2\max}$, after maximal short (40-m) sprint repetition in soccer players. Recently, Hazell et al. (11) demonstrated that 30-second near-maximal repetitions and 10-second all-out repetitions produced a similar significant $\dot{V}O_{2\max}$ improvement. The authors suggested that the generation of peak power output during the first few seconds of each training bout is more likely responsible for the adaptation than the total work completed. All together, these results and the results of this study may suggest that although performing long (30- to 45-second) sprint intervals is an effective aerobic training method, significant aerobic gain is possible even with short (6- to 8-second) sprint repetitions. In practical terms, because both training methods improved aerobic fitness equally, coaches can alternate between them to maintain player's training interest and enthusiasm during the busy soccer season.

In this study, we found a relatively large training-associated increase in predicted $\dot{V}O_{2\max}$. This may be related to the fact that the study was performed during the preseason period, when the players' initial fitness level was relatively low. Indeed, in previous studies, when sprint training was performed during the competitive season and players were in better physical shape, only a minor (3–6%) or non-significant increase in aerobic fitness was found (e.g., [8,13]).

The finding of this study is also important, because regardless of sport type, young athletes tend to avoid the traditional continuous endurance training. Therefore, short-sprint repetitions may be another training choice for aerobic fitness improvement in children and adolescents. This is especially true in team sports, where multiple fitness components, including speed and agility, are required.

In contrast to our hypothesis, SST induced similar beneficial changes in speed, agility, and anaerobic endurance compared with those of LST, whereas both training programs had no significant effect on jumping ability. The rationale for our hypothesis was based on the assumption that the short all-out 50-m sprints would be more effective in developing power and speed than the long submaximal speed 200-m repetitions, despite the matched total distance of both training programs. Short all-out sprint repetitions were previously found to enhance sprinting ability (see [6,15]). In contrast, Ferrari Bravo et al. (8) showed that 7 weeks of short (40-m) all-out sprint training was not associated with fitness improvement in junior soccer players. However, the training volume in their study was much lower than in this study (720 m twice a week compared with 800–1,200 3 times a week). Furthermore, as

previously mentioned, this study examined the improvement during the preseason period, when players' physical shape is relative low, whereas Ferrari Bravo et al. conducted their study during the competitive season, when players were in top shape. The relative low fitness level of the participants in this study may also be a reason for the unexpected significant improvement that was found in speed and agility among the LST group members. It is possible that in a low physical condition, nonmaximal speed repetitions (~85% of max speed) are sufficient to induce necessary metabolic and neuromuscular adaptations for speed and agility improvement.

In contrast to our hypothesis, no effect of training type, and no prechanges to postchanges were found in jumping ability. Although the lack of improvement in jumping ability for the LST group confirmed previous findings on junior soccer players after interval training (12), the absence of improvement in jumping ability for the SST was unexpected. Previous studies have shown that sprint training enhances jumping ability (6,15). However, these studies used a greater volume of training than did this study. When attempting to confirm the importance of training volume, Ferrari Bravo et al. (8) also failed to find a significant improvement in jumping after relatively low short-sprint training volume. It is also possible that skill and technical elements, important factors for jumping ability that were not practiced during the initial phases of the soccer preseason, interfered with the jumping results of this study, and masked improvement in muscle power among the participants. Given the importance of jumping ability to soccer, it is possible that additional strength training, such as weightlifting and plyometric exercise, might be required to improve muscular power. Although any type of specific strength training was not included in this study, the importance of strength training especially in adult soccer players is well documented in the scientific literature (23).

Although there were no statistically significant differences in fitness improvement between the 2 training programs, in competitive sports even a small, nonsignificant difference may distinguish between success and failure. For a relevant example, Cometti et al. (5) showed that French professional and amateur soccer players had a similar 30-m sprint performance, but the professionals had significantly lower 10-m lap times. The reported 10-m sprinting times ranged from 1.79 to 1.90 seconds (23). This means that the fastest players are on average about 1 m ahead of the slowest players after a sprint of only 10 m. This difference can influence the results of a game. Therefore, although not statistically different, the superiority of SST over LST in improving speed (30-m run: 2.49 vs. 1.55%, respectively) and agility (4×10 m: 2.90 vs. 1.04%, respectively) may suggest that it is still a preferred training method for these variables. Similarly, from a purely physiological perspective, long-sprint repetitions may be a preferred training method for aerobic fitness, as suggested from the different predicted $\dot{V}O_{2\max}$ changes (10.7 vs. 7.5% for the LST and the SST, respectively; Figure 1).

Finally, the aim of this study was to isolate the difference between the effect of the 2 sprint training protocols on specific fitness components. However, in a real and comprehensive soccer training, such isolation does not exist, and other fitness components such as strength and resistance drills play a major factor that improves the players' performance and assist in reducing injuries. Therefore, the effect of the 2 different training protocols should be studied in the future in "real-life" conditions (e.g., combined with training of other fitness components such as resistance training).

PRACTICAL APPLICATIONS

This study showed that long, near-maximal sprints, and short, all-out sprint training matched for total distance, are equally effective in enhancing aerobic and anaerobic fitness during the initial phases of the training season in youth soccer players. Therefore, during the busier soccer season, coaches can alternate between these methods to increase the players training enthusiasm.

Because even small differences may be important for competitive soccer, the nonsignificant superiority of SST over LST in improving speed and agility may suggest that SST is still a preferred training method for these variables. Similarly, LST may be a preferred training method for aerobic fitness.

Future studies should investigate the influence of other variables such as training volume, training period (preseason vs. busy competition schedule), the player's initial fitness level, and combination with training of other fitness components on fitness improvement. The influence of these training programs should also be examined on professional adult players with higher training and competitive demands.

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